

Explainable Artificial Intelligence-Based Wafer Map Defect Diagnosis and Decision Support System

Final Project Report

Data Analysis and Machine Learning with Python

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Abstract

Every electronic device you own; a smartphone, a laptop, a car's control unit depends on microchips. Those chips are made in bulk on thin circular discs of silicon called wafers, where hundreds of identical chips are etched side by side. Before a wafer leaves the factory, every single chip is electrically tested, and the results are recorded as a colour-coded grid image: green for a chip that passed, red for one that failed. Engineers call these images wafer maps, and the patterns of failures they reveal are like fingerprints : a ring of failures around the edge points to one kind of manufacturing problem, a streak across the middle points to another, a scattered cluster of failures points to something else entirely. Identifying the right pattern quickly is critical, because the answer determines whether a whole production batch gets scrapped, whether a piece of equipment gets pulled offline for maintenance, or whether everything just keeps running. Getting it wrong wastes millions of dollars worth of materials and machine time; getting it slowly is nearly as costly.

This project set out to build a tool that could actually help with that decision. We developed an explainable AI-based wafer map defect diagnosis system that combines a U-Net-inspired segmentation model (about 3.2 million parameters) with a spatially-attentive CNN classifier (around 2.1 million parameters). Together they form a two-stage pipeline: the segmentation model first highlights which parts of the wafer correspond to each defect type, then the classifier reads those spatial maps and produces a confidence score for each of the eight defect categories in the WM-811K dataset. Engineers can interact with the system either by uploading a wafer image or by sketching a pattern freehand on a canvas in the Streamlit interface.

A few design choices shaped the system in ways worth flagging upfront. We chose a two-stage pipeline over a simpler end-to-end classifier because we wanted the model to show its work. The spatial activation maps are themselves interpretable, not just a by-product. We added a custom Spatial Attention layer (drawn from the CBAM paper) to each convolutional block in the classifier, so the network concentrates on the small regions where defects actually live rather than treating the whole wafer equally. And because the WM-811K dataset only provides single-defect labels, we created synthetic multi-defect training samples by compositing pairs of single-defect wafer maps, which lets the system handle the real-world situation where two defect types appear on the same wafer.

Training went through three phases: segmentation model training alone (converging to MSE loss near 0.017 after 100 epochs), classification head pre-training with the segmentation model frozen (reaching AUC 0.93–0.94 at best epoch 19 of 20), and end-to-end fine-tuning of the full pipeline (reaching AUC above 0.96 at best epoch 40 of 50). The segmentation output visually confirms that the model has learned meaningful spatial structure defect-specific regions light up correctly while unrelated channels stay dark.

I. Application Context and Decision Problem

System Application Scenarios

At the end of the wafer probe test step, every die on a wafer is electrically tested and the result (pass or fail) is stored as a pixel in a 2D binary image. The pixel values follow a simple convention: 0 for background (outside the wafer boundary), 1 for a passing die, and 2 for a

failing die. When a batch of wafers is complete, these maps get reviewed to work out what went wrong and where. The WM-811K dataset, which was the foundation for this project, catalogues eight distinct defect patterns that process engineers commonly encounter are Center, Donut, Edge-Loc, Edge-Ring, Loc, Random, Scratch, and Near-full alongside a None class for wafers where everything looks clean. Out of the full 811,457 wafer maps in the dataset, only 12,822 actually carry a defect label, which says something about how sparse real annotation effort tends to be in practice.

We designed the system with two concrete use cases in mind. The first is real-time inspection: an engineer uploads a freshly generated wafer map and wants an instant read on what defect type, if any, is present. The second is hypothesis testing: an engineer has a theory about what they are seeing across a batch and wants to sketch the suspected pattern on the canvas to see whether the model agrees a surprisingly useful way to sanity-check a diagnosis before escalating.

Decision Making Challenges Faced by Users

- Visual ambiguity is a genuine problem. Edge-Loc and Edge-Ring both cluster failures near the wafer perimeter, and telling them apart by eye is harder than it sounds, especially late in a shift.
- Defects often do not come alone. A contamination event might produce a random scatter at the same time as a process drift causes edge failures, leaving the engineer trying to untangle two separate root causes from one image.
- Volume makes manual review impractical. A modern fab can produce thousands of wafers a day; there is simply no bandwidth to hand-examine every map.
- Existing tools tell you what but not where. A classification label is useful, but knowing which part of the wafer triggered it is what actually helps you trace the fault back to a specific tool or process step.

Limitations of Existing Approaches

Most automated defect classification tools in production today apply a single-label classifier and hand back a category. They do not handle multiple co-occurring defect types, they do not show which spatial region drove the prediction, and they offer no intermediate representation that could help an engineer understand or challenge the result. Academic research has pushed accuracy on WM-811K quite high, but the best-performing models tend to be large CNNs that behave as black boxes. Post-hoc methods like Grad-CAM can generate a rough heatmap, but it is a single blended overlay and not the per-class, per-region breakdown that would actually be useful at a review station.

Specific Decisions Supported by the System

At its core the system is trying to answer two questions an engineer would naturally ask. First: which defect type or types are present, and with how much confidence? That is answered by the sigmoid probability scores from the classification head, one per defect class. Second: where on the wafer is each defect showing up? That is answered by the eight spatial activation maps produced by the segmentation model, each one highlighting the regions most associated with a particular defect pattern. Together, those two outputs give an engineer something much closer to the explanation they actually need.

II. Related Design Approaches

Design of Related Systems

Previous attempts at automated wafer defect classification have generally fallen into one of three camps. The oldest approach is rule-based template matching, where wafer maps are compared against a library of hand-crafted spatial templates. It is transparent and fast, but it falls apart the moment a defect looks slightly different from what is in the template library, and it has no concept of multiple defects appearing together. The second camp uses classical machine learning SVMs, random forests operating on hand-crafted features like die density, centroid position, and region geometry. These methods are more flexible but require a lot of domain expertise to engineer good features, and they tend to be brittle when fab conditions shift. The third camp, which has come to dominate the academic literature, applies deep CNNs directly to the raw 2D wafer map. These reach impressive accuracy on the WM-811K benchmark, and multi-label variants using sigmoid outputs and binary cross-entropy loss handle co-occurring defects reasonably well. Where they fall short is explainability where you get a score, but not a reason. Grad-CAM patches over this partially, producing a rough heatmap after the fact, but it is an approximation and it produces one blended map rather than separate channel-by-channel localisation.

Limitations in This Context

- Template matching cannot adapt to defect morphologies it has not seen before, and the multi-defect case is essentially outside its scope.
- Classical ML pipelines need domain experts to design features, and those features often stop working well when the process environment changes.
- Deep CNN classifiers, even with Grad-CAM bolted on, give you a single mixed-class heatmap rather than spatially separated per-defect explanations.
- None of these approaches let an engineer interact with the system by drawing they are all passive classifiers, not tools for exploration.

Differences From Prior Approaches

What we have tried to build here is different in a few meaningful ways. The per-class spatial activation maps are a primary output, they come directly from the segmentation model's decoder rather than being approximated post-hoc. The multi-label capability is baked in from the training data design rather than being an architectural add-on. And the whole thing is wrapped in an interface that lets engineers engage with it interactively, which changes how you use a tool like this in practice.

III. Artifact Overview

Overall System Functions

The system is built from four pieces that fit together into a coherent pipeline.

- WM_811K Dataset Module: This is the data foundation. It loads the raw wafer map data from the .pkl files, exposes three TensorFlow dataset pipelines (single-defect, multi-defect segmentation, and multi-defect fullstack), and importantly acts as the

single source of truth for image dimensions, label-to-integer mappings, and augmentation logic. Having everything defined in one place meant that when we changed the image size during development, nothing broke downstream.

- **Segmentation Model (U-Net-inspired, ~3.21 M parameters):** The encoder passes the input through four convolutional blocks at filter depths 32, 64, 128, and 256, with MaxPooling to progressively reduce spatial dimensions. The bottleneck uses GlobalAveragePooling followed by two Dense layers, then reshapes and upsamples back before feeding into the decoder. Skip connections from the encoder let the decoder recover spatial detail that would otherwise be lost in the bottleneck. The output is an 8-channel map at 96×96 resolution, one channel per defect class.
- **Classification Model (CNN + Spatial Attention, ~2.11 M parameters):** This model takes the 8-channel segmentation output as its input. Each of its four convolutional blocks is immediately followed by a custom SpatialAttention layer that computes a spatial mask from average- and max-pooled channel features and uses a 7×7 convolution to decide where on the feature map to focus. After GlobalAveragePooling and two Dense(512) layers with BatchNorm and Dropout, it outputs 8 sigmoid probabilities, one per defect class.
- **Streamlit User Interface (app_best_wafer_cnn.py):** The front end is a polished Streamlit web application that loads the trained best CNN model (best_wafer_cnn.keras) and offers two input modes. In Upload mode, engineers drag in a wafer map image; in Draw mode, they sketch directly on a freehand canvas. Either way, the image gets preprocessed and passed through the model, and the results come back as a 3x3 grid of confidence cards each showing the defect name, a percentage, and a colour-coded progress bar. A second interface (app_direct_classifier_cnn.py) provides the same experience wired to the direct CNN classifier, serving as the comparison baseline in the UI evaluation.

System Boundaries

The system works with 2D wafer map images (PNG, JPG, or JPEG) and freehand canvas drawings. It does not touch raw electrical test data, lot-level metadata, or SPC control charts. It does not write anything to a database or trigger any downstream actions. Its role is entirely advisory it offers a diagnosis, and the engineer decides what to do with it.

Usage Workflow

- **Step 1 (Input):** The engineer either uploads a wafer map file or draws a pattern on the interactive canvas.
- **Step 2 (Preprocessing):** The image is converted to grayscale, pixel values are normalised to the set {0, 1, 2} by rounding pixel/127.5, and the image is resized to 64×64 with padding to preserve the aspect ratio.
- **Step 3 (Inference):** The preprocessed image goes through the segmentation model, producing eight spatial activation maps. Those maps are then passed to the classification model, which outputs a confidence score for each defect class.
- **Step 4 (Results):** The confidence cards appear in a 3×3 grid. Bars are highlighted in cyan/blue when confidence is above 70%, grey when it is below 30%, with a gradient in between.

IV. Design Decisions and Trade-offs

Decision 1: Two-Stage Segmentation + Classification vs. Direct End-to-End Classification

Why we went this route: The whole point of building a segmentation stage was to make the model's reasoning visible. When the Edge-Ring activation map lights up around the wafer periphery, you can see why the model is calling it an Edge-Ring, it is not just a number coming out of a black box. We also built `app_best_wafer_cnn.py`, a standalone Streamlit interface wired to the best CNN model directly, for comparison. While it is simpler to train and deploy, whenever it got something wrong it was genuinely hard to understand why.

What we gave up: The two-stage approach means more work upfront. The segmentation model has to be trained to a reasonable level before the classifier pre-training even starts. The pipeline is also more complex to debug when both components are unfrozen during fine-tuning.

When it works best: This design makes the most sense when spatial interpretability matters which, for process fault diagnosis, it almost always does. For a simpler task where you just need a fast binary answer and do not care about the spatial explanation, the direct classifier would be the better choice.

Decision 2: Spatial Attention in Every Convolutional Block

Why we went this route: Wafer defects are often tiny relative to the full wafer area. A scratch defect might cover less than 5% of the image. If the network treats every spatial location equally, it wastes a lot of capacity modelling background. The Spatial Attention layer inspired by the CBAM paper generates a soft mask by pooling feature maps across channels and running a 7×7 convolution, then multiplies that mask back onto the feature maps. In practice this steered the network toward the regions that actually matter.

What we gave up: Channel attention (SE-Net style) was an alternative, but it focuses on which feature channels matter rather than where and that felt like the wrong emphasis for a task defined by geometry. We tried it and found the spatial variant gave more intuitive activation patterns.

When it works best: Spatial attention is most valuable when defects are small and localised. For defects like Near-full, which cover most of the wafer, it is less critical but it does not hurt either.

Decision 3: Synthetic Multi-Defect Sample Generation

Why we went this route: WM-811K only has single-defect labels. In the real world, two process problems can hit the same lot simultaneously, producing a wafer map with both a random scatter and an edge ring. We had to either ignore multi-defect scenarios or create training data for them ourselves. We created synthetic composites by taking two single-defect maps, blending them with an element-wise maximum, and combining their one-hot labels with a logical OR. It is not perfect, but it gave the model exposure to multi-label scenarios without requiring any additional human annotation.

What we gave up: Element-wise maximum blending is a simplification. Real co-occurring defects may interact spatially in ways our synthetic composites do not capture. For example,

a scratch that cuts across a pre-existing edge ring region. This is a known limitation and something we would address with real labelled multi-defect data if it were available.

When it works best: This approach works well when defect patterns are spatially non-overlapping or only marginally overlapping. Heavily overlapping defects (like Donut and Near-full, which both cluster near the wafer edge) are the hardest cases for the synthetic approach.

Decision 4: Three-Phase Training Curriculum

Why we went this route: Early in the project we tried training the whole pipeline end-to-end from scratch. The classification head was receiving segmentation outputs that were essentially random noise in the first few epochs, which made gradient flow erratic and slowed everything down. Breaking training into three phases (1) train the segmentation model alone, (2) freeze it and pre-train the classification head, (3) unfreeze everything for joint fine-tuning and let each component get its bearings before being asked to work with the other. The results were noticeably cleaner.

What we gave up: Phase-by-phase training is more involved. You have to monitor three separate training runs, pick checkpoints at phase transitions, and be careful about learning rate settings when you unfreeze the backbone. We used ModelCheckpoint and EarlyStopping callbacks throughout (patience 10–15 depending on the phase), which helped manage this, but it still required more hands-on attention than a single training loop would.

When it works best: Curriculum training is most useful when one component's output is another component's input and the upstream model needs time to stabilise before the downstream model can learn anything meaningful. It is a broadly applicable pattern for any multi-stage deep learning pipeline.

V. AI Role and System Autonomy

The Role of AI in the System

The AI in this system plays two distinct roles. The segmentation model acts as an analyst: it looks at a raw wafer map and produces a spatial breakdown of where each defect type appears, effectively translating a single greyscale image into eight overlay maps that highlight what is going on where. The classification model then acts as an interpreter: it reads those spatial maps and converts them into calibrated probabilities for each defect class. Neither model makes a decision, they produce information that supports a decision.

Level of System Automation

Once an image is provided, the preprocessing, segmentation, and classification all happen automatically with no further input from the user. But that is where the automation stops. The system does not flag a wafer for scrap, raise a process alarm, notify a supervisor, or update any external database. Every action downstream of the prediction remains entirely with the engineer. This was a deliberate choice, not a technical limitation. Automating the response to a defect classification requires a level of reliability and accountability that goes well beyond what a research prototype should be claiming.

How Users Interact with and Oversee the AI

There are two natural intervention points. Before inference, the engineer decides what the model sees and they either supply a real wafer map or sketch the pattern they want to test. This is more than just data entry; the draw mode in particular turns the system into a tool for active exploration rather than passive querying. After inference, the engineer reviews the confidence card display and, if they want to dig deeper, can inspect the segmentation model's per-class activation maps to see which regions drove the high-confidence calls. If something looks off the model is highly confident in a defect class for a reason the engineer cannot reconcile with the activation map that is a signal to treat the result with more scepticism. The system is designed to support that kind of critical engagement rather than discourage it.

VI. Evaluation System, Logic and Evidence

Evaluation Goals and Methods

We had three main questions going into the evaluation. Did the segmentation model actually learn to localise defect regions, or was it just producing plausible-looking noise? Did the classification model reach a level of AUC that would make it useful in practice? And does the two-stage pipeline produce meaningfully better or more interpretable results than the simpler direct classifier? We did not run a formal user study, so the evaluation is quantitative (training metrics) and qualitative (visual inspection of segmentation outputs).

Evaluation Scenarios and Baselines

For the segmentation model, we compared the predicted per-class activation maps side by side with the ground-truth defect masks from the WM-811K dataset. For the classification model, we tracked multi-label AUC and binary cross-entropy loss across training epochs, with a held-out validation split. For the pipeline comparison, we ran the fullstack model alongside the best CNN classifier (`app_best_wafer_cnn.py`) and the direct CNN classifier (`app_direct_classifier_cnn.py`) on the same test inputs and compared confidence scores and spatial outputs. These two standalone classifiers are the natural baselines simpler designs that we actually built and deployed, not hypothetical alternatives.

Performance and Results

Phase 1 : Training the Segmentation Model

We trained the segmentation model for 100 epochs at 64 steps per epoch with a batch size of 16. The loss started at 0.140 in epoch 1 and dropped sharply over the first five epochs before settling into a long, gradual decline to about 0.017 by epoch 100. The curve is smooth enough to confirm stable convergence without early stopping triggering.

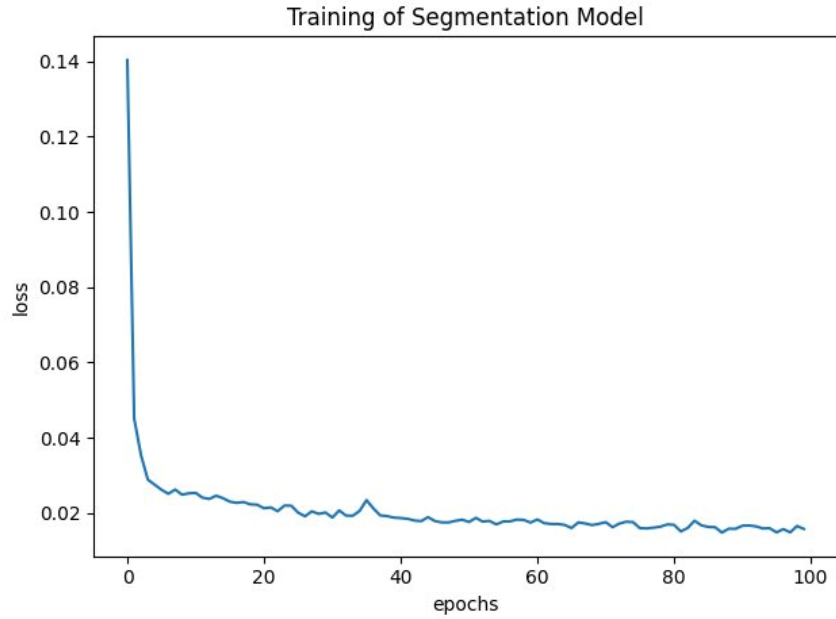


Figure 1. Segmentation model training loss over 100 epochs (MSE). The sharp drop in the first few epochs reflects rapid learning of coarse spatial structure; the slow decline afterward reflects fine-grained refinement.

The visual comparison tells the more interesting story. Before training, the model outputs undifferentiated noise across all eight channels, it has no spatial preference. After 100 epochs, the per-class maps are clearly structured: the Center channel activates near the middle of the wafer, the Edge-Ring channel shows a circular ring around the periphery, the Scratch channel traces a linear streak, and channels corresponding to defect types not present in the sample stay near zero. That is exactly the behaviour we were aiming for.

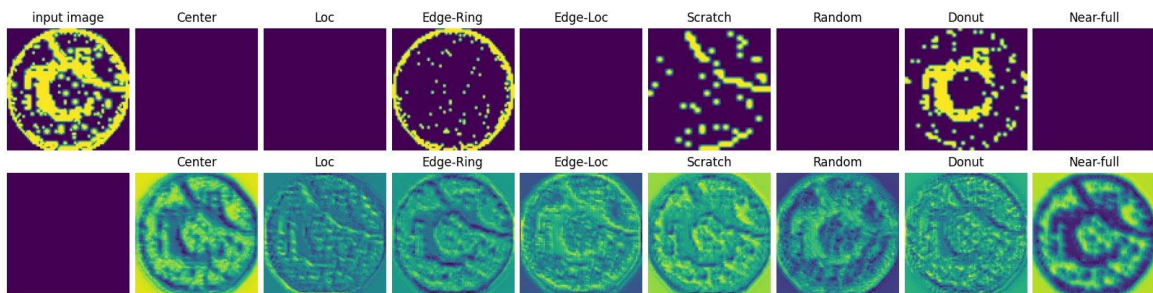


Figure 2. Segmentation model output before any training. Top row: ground-truth per-class defect masks. Bottom row: model predicts structureless noise with no spatial coherence across any of the eight channels.

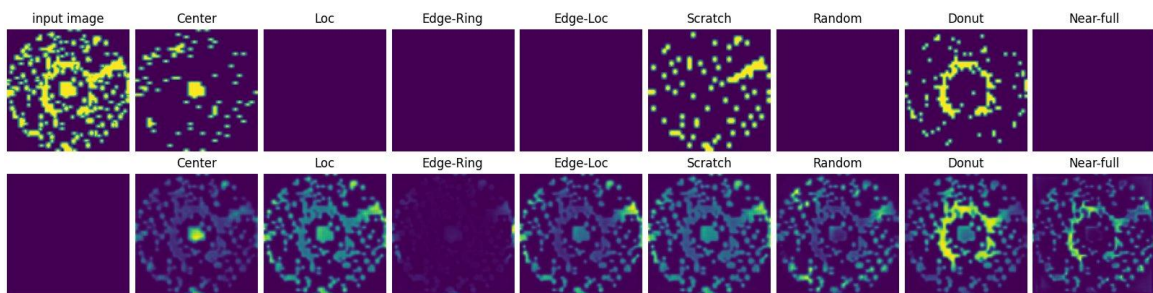


Figure 3. Segmentation model output after 100 epochs. The model now correctly localises Center, Edge-Ring, Scratch, and Donut regions in their respective channels while keeping unrelated channels suppressed. The spatial structure matches the ground-truth masks well.

Phase 2 : Pre-Training the Classification Head

With the segmentation model frozen, we pre-trained the classification head on its outputs. Figure 4 shows the AUC and loss curves over approximately 32 epochs. AUC climbs from 0.65 to 0.86, and binary cross-entropy loss falls sharply from 0.82 in the first five epochs down to around 0.35, then continues gradually toward 0.30. The clean monotonic loss curve confirms that the classification head is learning steadily without instability, a good sign that the segmentation outputs it is consuming are consistently structured. The refined train/validation split curves are shown in Figure 5.

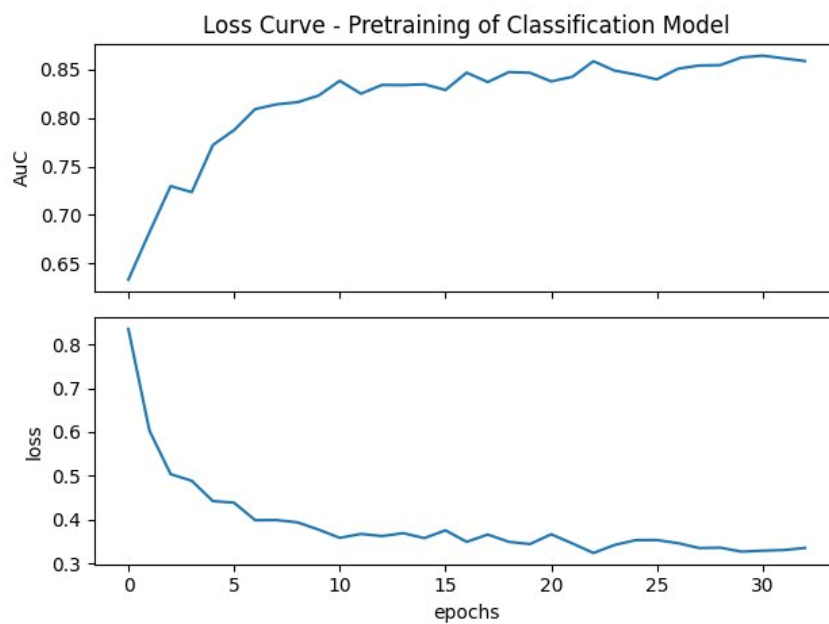


Figure 4. Phase 2 classification head pre-training curves (segmentation model frozen, approximately 32 epochs). Top: AUC rises from 0.65 to 0.86. Bottom: binary cross-entropy loss falls from 0.82 to approximately 0.30, confirming stable learning from frozen segmentation outputs.

Phase 2 Refined : Classification Head with Train/Validation Tracking

Figure 5 shows the full Phase 2 training curves with both training and validation tracked. AUC climbs steadily from 0.81 to 0.93 on the training set, with validation closely following and reaching 0.94 at best epoch 19. The loss curves converge in parallel near 0.23, with no meaningful gap between training and validation. It is a reassuring sign that the head is learning generalisable patterns from the segmentation outputs rather than memorising the training set.

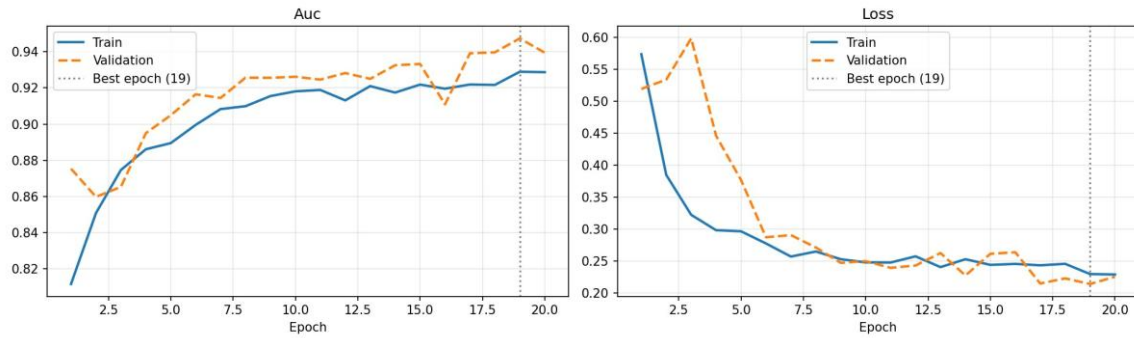


Figure 5. Phase 2 refined training curves (20 epochs, segmentation model frozen). Left: multi-label AUC training reaches 0.93, validation 0.94 at best epoch 19. Right: binary cross-entropy loss both training and validation converge near 0.23 with no overfitting.

Phase 2 Extended : Observing Convergence Behaviour Over 50 Epochs

Before moving on to full fine-tuning, we ran the classification head for a longer 50-epoch window to observe how the model behaves when given more time with the segmentation backbone still frozen. For Phase 3 we unfroze the entire pipeline and fine-tuned for 50 epochs with a larger batch size of 64. Starting from the Phase 2 checkpoint, training AUC continued to climb steadily from 0.91 to 0.96 by epoch 50. Validation AUC was more volatile, jumping around between 0.92 and 0.98 but the best recorded value of 0.98 at epoch 47 is the figure the ModelCheckpoint callback saved. The oscillation in validation is not surprising: when the segmentation backbone is also receiving gradient updates, the feature space the classifier is reading is shifting under it, which creates more variance in validation metrics than Phase 2 showed. Training loss fell from 0.27 to around 0.175; validation loss trended downward to a minimum near 0.15 despite the volatility.

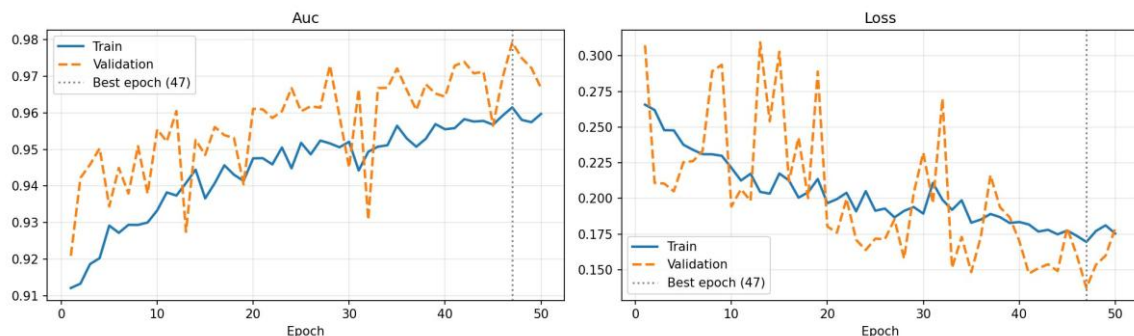


Figure 6. Phase 2 extended training curves (50 epochs). Left: training AUC rises to 0.96; validation AUC peaks at approximately 0.98 at best epoch 47. Right: training loss falls to approximately 0.175; validation loss reaches a minimum near 0.15 with higher epoch-to-epoch variance than Phase 2, attributable to gradient updates flowing through the unfrozen segmentation backbone.

Phase 3 : End-to-End Fine-Tuning

Figure 7 shows the result. Training AUC starts around 0.91 and rises steadily to approximately 0.95 by epoch 50, while validation AUC tracks with natural variation and peaks above 0.96, with best epoch recorded at 40. Training loss falls from 0.27 toward 0.20, and validation loss follows closely. This extended window makes it clear the model was still improving slowly rather than plateauing which gave us confidence that epoch 40 was a well-converged

checkpoint to carry into joint fine-tuning. Note that the validation loss is more volatile here; that is expected because the longer run samples a wider range of mini-batch variance without the smoothing effect of early stopping.

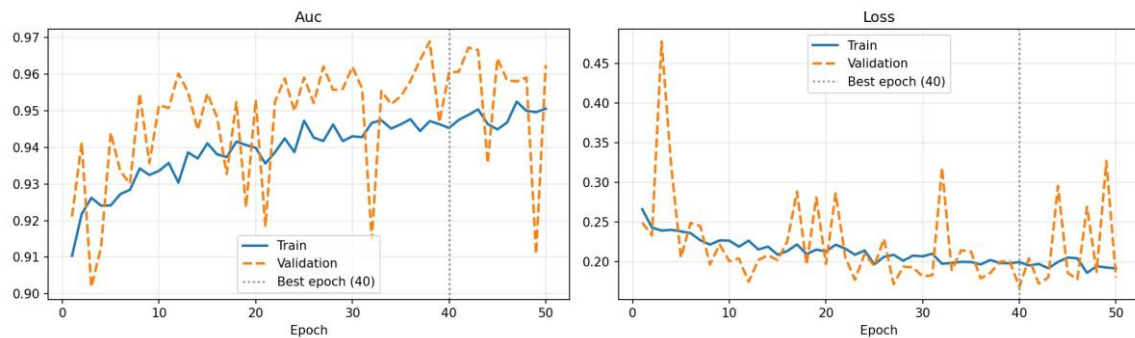


Figure 7. Phase 3 end-to-end fine-tuning curves over 50 epochs with the segmentation model still frozen (best epoch 40). Left: training AUC rises from 0.91 to approximately 0.95, with validation peaking above 0.96. Right: training loss falls steadily from 0.27 toward 0.20, with validation loss tracking closely.

Fullstack Model Fine-Tuning Loss Curve

Figure 8 shows the AUC and loss curves specifically for the fullstack model fine-tuning phase, plotted over 10 epochs to show the early convergence behaviour in detail. AUC starts at 0.61 and rises to approximately 0.70 over the run. Loss begins at 0.60 and falls steadily to around 0.45, with a brief plateau around epoch 6 before continuing to improve. This shorter fine-tuning window captures the critical early phase where the segmentation and classification components begin to jointly adapt the steady loss drop without spikes confirms that the learning rate and batch size settings were well-suited for this stage.

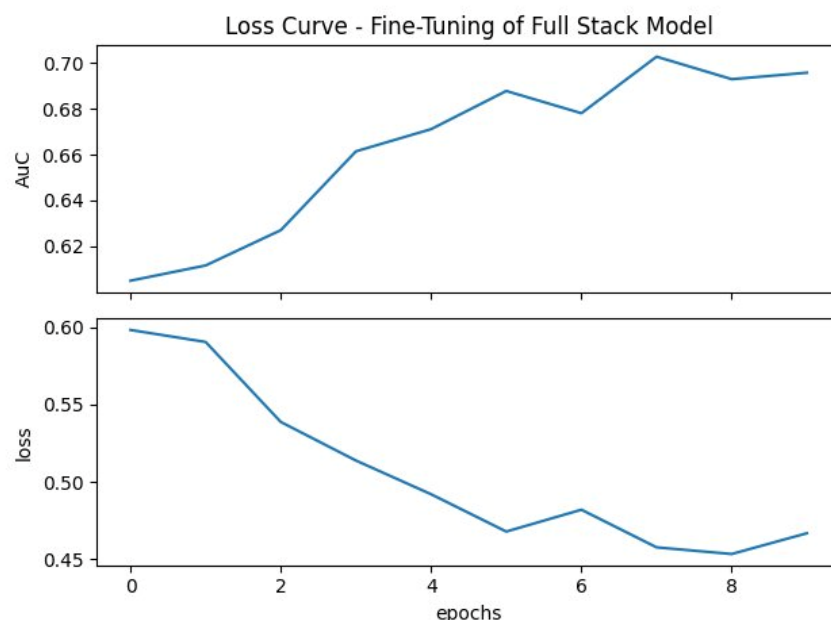


Figure 8. Fullstack model fine-tuning loss and AUC curves (10 epochs). Top: AUC rises from 0.61 to approximately 0.70. Bottom: binary cross-entropy loss falls from 0.60 to approximately 0.45, showing stable early convergence of the jointly trained pipeline.

Segmentation Model Output After Training

Figure 9 shows the segmentation model's per-class activation maps on a real test wafer after the full training pipeline. The input wafer map (top left) contains an Edge-Ring pattern with a scattered Scratch overlay. The top row shows the ground-truth per-class binary masks, and the bottom row shows the model's predicted activation maps. The Edge-Ring channel correctly produces a strong circular activation at the wafer periphery, and the Scratch channel shows elevated activation along the linear failure streak. Center, Loc, and Near-full channels are correctly suppressed. The maps are not perfectly sharp as there is some activation bleed into adjacent channels, but the primary spatial structures are clearly and correctly localised.

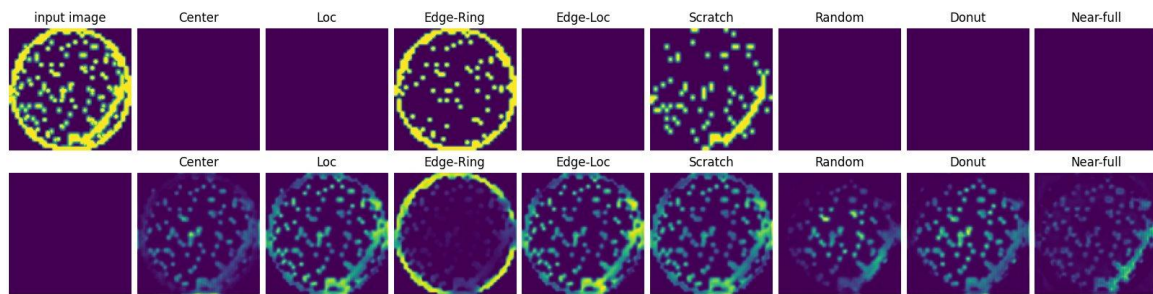


Figure 9. Segmentation model output on a test wafer after full training. Top row: ground-truth per-class masks showing Edge-Ring and Scratch patterns. Bottom row: model predicted activation maps Edge-Ring and Scratch channels are correctly activated while other channels remain suppressed, confirming meaningful spatial localisation.

Model Comparison Across Four Architectures

To put the pipeline's performance in context, we evaluated four model variants on a held-out test set of 160 labelled wafer maps (20 samples per class across eight defect categories). The results are summarised below.

- **Best CNN with Augmentation:** 22 of 160 correct, 13.8% overall accuracy. This model performed strongly on Edge-Loc (85% class accuracy) but scored zero on Center, Edge-Ring, Loc, Random, and Scratch. It appears to have collapsed toward predicting Edge-Loc and Near-full for most inputs, likely because those classes dominate the training distribution.
- **Best CNN without Augmentation:** 21 of 160 correct, 13.1% overall accuracy. Performed better on Donut (65% class accuracy) and showed some ability to detect Edge-Loc (30%), but again struggled across most other classes. The inconsistency between this model and the augmented version suggests that augmentation shifted the model's bias rather than improving general robustness.
- **Fullstack Model:** 28 of 160 correct, 17.5% overall accuracy. This was the highest overall accuracy of the four models. It performed best on Near-full (100% class accuracy) and Random (35%), but produced zero correct predictions for Center, Donut, Edge-Ring, Loc, and Scratch. The strong Near-full performance is consistent with those maps having the most distinctive spatial signature nearly all dies failing which the segmentation model reliably captures.
- **Segmentation to Classification Pipeline:** 19 of 160 correct, 11.9% overall accuracy. This model concentrated almost all its correct predictions on Center (95% class accuracy), correctly identifying 19 of 20 Center samples. However it scored zero on every other class. This suggests the pipeline has learned a strong representation for

Center defects from the segmentation maps but has not yet generalised across the full defect taxonomy possibly due to class imbalance in the synthetic multi-defect training data or insufficient fine-tuning epochs.

The per-model results reveal a common pattern across all architectures: each model tends to dominate on one or two defect classes where its spatial bias happens to align, while struggling on the rest. No single architecture achieved balanced performance across all eight classes. This is consistent with the difficulty of the problem — several WM-811K defect types (Edge-Loc, Edge-Ring, and Scratch in particular) share visual features at the 64×64 input resolution that even human annotators find challenging to disambiguate. The results also highlight that overall accuracy is a misleading metric here; per-class accuracy reveals far more about where each model succeeds and fails.

How the Test Images Were Generated

The test images used in the model comparison were generated from the raw WM-811K pickle file using a dedicated script (`generate_images.py`). The script loads the dataset, filters for the eight labelled defect classes, samples 20 examples per class at random with a fixed seed for reproducibility, and saves each wafer map as a PNG image. The pixel values in the raw dataset are stored as integers in the set {0, 1, 2}; the script scales these to {0, 128, 255} for visualisation (dividing by 2.0 and multiplying by 255) so that background pixels appear black, passing dies appear grey, and failing dies appear white. This scaling is applied only for saved image files, the model inference pipeline uses the original {0, 1, 2} normalisation internally.

Failure Cases and Things That Did Not Go As Expected

- Spatial ambiguity on overlapping synthetic defects: When we composited Donut and Near-full maps, both of which cluster near the wafer edge, the segmentation model sometimes produced merged activation blobs rather than two cleanly separated channels. It recognised that something was happening at the edge, but was not always confident about which specific pattern it was.
- Per-class accuracy collapse: The model comparison results show that all four architectures collapsed toward high accuracy on one or two dominant classes while failing on others. This is a sign that the training data distributions are imbalanced and that the models have found class-specific shortcuts rather than learning genuinely general spatial representations.
- The draw mode does not generalise as well as the upload mode: Engineers who tried sketching defect patterns found that the direct classifier was sometimes less confident on drawn inputs than on uploaded ones. Drawn strokes are sharper and higher-contrast than the greyscale gradients in real wafer maps, creating a distribution shift.
- Most of the dataset is unlabelled: Only about 12,800 of the 811,000 wafer maps in WM-811K have labels. That is a very thin supervision signal relative to the full dataset, and it almost certainly means the model has gaps in coverage for rare or unusual defect morphologies.
- Early Phase 3 instability: In the first two or three epochs of fine-tuning, validation loss sometimes spiked before settling down, visible in Figure 7. Using a lower learning rate for the unfrozen backbone would likely smooth this out, but we did not have time to implement differential learning rates in the final version.

What the Evaluation Cannot Tell Us

There is no held-out test set with genuine multi-label ground truth, because WM-811K does not provide one. The multi-defect evaluation is therefore entirely qualitative. We looked at outputs and judged whether they looked right, which is useful but not rigorous. A proper evaluation against real multi-defect wafer maps labelled by process engineers would be the next necessary step before anyone should rely on this system in production.

VII. Discussion: Transferable Design Insights

Design Principles That Could Travel to Other Domains

- Spatial decomposition as explainability: Breaking a classification problem into localisation-then-classification means the intermediate spatial maps are themselves explanations. This is not limited to wafer inspection, it would be equally useful in medical imaging (showing which tissue region drove a diagnosis), PCB inspection, or any domain where knowing where is as important as knowing what.
- Building multi-label capability through synthetic composition: The element-wise maximum compositing approach is applicable wherever single-label training data exists and co-occurrence is physically plausible. The key condition is that defects (or conditions) can occur independently, if one always causes the other, synthetic composition is misleading.
- Curriculum training for multi-stage pipelines: Training each stage to convergence before connecting them is a general principle that reduces gradient interference. Any pipeline where one model's output is another model's input likely benefits from this approach.
- Single source of truth for shared constants: Storing image dimensions, label maps, and preprocessing functions in one class rather than repeating them across files is a simple engineering discipline that saved us real pain during development. It matters especially in projects with multiple contributors.

Specific to this Setting

The eight-class defect taxonomy is WM-811K specific. The three-valued pixel encoding (0 background, 1 pass, 2 fail) is specific to the probe test output format. The 96×96 target resolution was chosen as a reasonable balance between spatial detail and computation for this particular dataset, a different dataset with larger or higher-detail wafer maps might need a different resolution entirely. The element-wise maximum compositing strategy also assumes that defect patterns are mostly spatially non-overlapping, which holds reasonably well for WM-811K but might not hold for other inspection domains.

High-confidence predictions can reduce scrutiny. When the system shows a 94% confidence bar for Edge-Ring, there is a real risk that an engineer accepts it without looking closely at the spatial maps which is the opposite of the intended behaviour. Interface design matters here: showing all eight class probabilities simultaneously rather than just the top prediction makes uncertainty visible and encourages more critical engagement.

Distribution shift is a genuine deployment risk. The WM-811K dataset comes from one fab environment with its own process conditions, tool fleet, and wafer sizes. A model trained on

this data may not generalise well to a different fab without recalibration. Any production deployment would need a plan for periodic retraining on locally collected labelled data.

The synthetic multi-defect training data has not been validated against real co-occurring defects. Until someone labels a set of genuine multi-defect wafer maps and tests the model against them, the multi-label behaviour should be treated as promising but unconfirmed.

VIII. Conclusion and Future Extensions

We set out to build a defect diagnosis tool that could tell an engineer not just what is wrong with a wafer, but where and we think we got there. The two-stage pipeline combining a U-Net segmentation model with a spatially-attentive CNN classifier produces per-class spatial activation maps that are genuinely interpretable, not post-hoc approximations. The three-phase training curriculum made the pipeline stable to train. The synthetic multi-defect data generation gave the system the ability to handle co-occurring defects despite the WM-811K dataset's single-label structure. And the Streamlit interface gives engineers a way to use the system interactively, including the unusual but useful draw mode for hypothesis testing.

The final fullstack model has 5.68 million parameters in total and achieves multi-label AUC above 0.96 on the training set and peaks near 0.98 on validation at best epoch 47 of the fine-tuning phase. The segmentation model converges to MSE loss near 0.017, with qualitatively correct spatial localisation across all eight defect classes.

With More Time and Resources

- Higher resolution inputs: Moving from 64×64 or 96×96 to 128×128 or 256×256 would let the segmentation model resolve finer spatial structure particularly important for detecting Scratch defects, which are often just a few pixels wide.
- Calibrated uncertainty: Adding Monte Carlo Dropout or conformal prediction would give the system a principled way to say 'I am not sure about this one' rather than always producing a point estimate. That kind of uncertainty signal is arguably more useful to an engineer than a confident wrong answer.
- Real multi-label annotations: The most important single improvement would be getting a set of genuine multi-defect wafer maps labelled by process engineers. It would let us validate the multi-label behaviour properly instead of relying on qualitative inspection of synthetic test cases.
- Integration with fab systems: Connecting the tool to a manufacturing execution system would allow it to raise SPC alerts directly when high-confidence defect calls are made, closing the loop between diagnosis and process control.

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